IoT Fundamentals & Sensor Simulation

Mission 1 – IoT Architecture, Protocols, Sensor Simulation and Visualization

1. Objective

To understand the basic architecture of an Internet of Things (IoT) system and simulate sensor data using Python. The project generates temperature and humidity readings, stores them in CSV format, and visualizes the readings using Matplotlib.

2. IoT Architecture

The proposed architecture consists of sensing, network/communication, processing/storage, and application layers. Sensors generate physical-world data. A gateway or network connects the sensing devices to the backend. Cloud or server-side processing can store and analyze the data. The application layer presents useful information to the user.

3. Main IoT Layers

Perception/Sensing Layer: collects data using sensors such as temperature and humidity sensors. Network Layer: transfers sensor data using communication technologies and protocols. Processing/Cloud Layer: stores, processes, and analyzes the collected data. Application Layer: displays the processed information through dashboards or applications.

4. Communication Protocols

MQTT is a lightweight publish/subscribe messaging protocol commonly used in IoT because it has low communication overhead. HTTP follows a request/response model and is widely used for web-based communication and REST APIs. In a practical IoT system, either protocol can be selected according to application requirements.

5. Sensor Simulation

Because physical sensors are not required for this beginner simulation, Python is used to generate representative temperature and humidity readings. The generated values are stored in sensor_data.csv for further analysis.

6. Data Visualization

Matplotlib is used to plot the simulated readings against the reading number. The resulting graph helps observe changes in temperature and humidity over the simulated measurement period.

7. Technologies Used

Python, CSV, Matplotlib, MQTT/HTTP concepts, and basic IoT architecture concepts.

8. Workflow

Sensor data generation → Data collection → CSV storage → Data visualization → Analysis/application.

9. Result

A working Python-based sensor simulation was developed. It generates 30 simulated temperature and humidity readings, saves them as CSV data, and produces a visualization graph.

10. Conclusion

This mission demonstrates the basic flow of an IoT data system from sensing and communication concepts to data collection and visualization. The same architecture can be extended later with real sensors, an MQTT broker, cloud storage, and a real-time dashboard.

Sample Data

Reading	Temperature (°C)	Humidity (%)
1	27.44	69.65
2	27.50	69.73
3	28.08	70.26
4	28.37	68.15
5	28.14	67.22
6	28.01	67.51
7	27.81	65.90
8	28.11	65.90
9	27.64	65.25
10	27.86	63.21